

Cognitive Abstraction Layers (CAL) — Pre-paper v1.4

Cognitive Abstraction Layers (CAL): A Research Architecture for Hierarchical Semantic Compression in AI Systems

Abstract

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Research repositories:

Layer	Repo	Paper
CAL (framework)	<u>Cognitive- Abstraction- Layer-CAL</u>	This document
	<u>TENSOR- BASED-</u>	
L2 — TCO	<u>COGNITIVE- OVERSIGHT- TCO</u>	L2/Documentacion/TCO_Paper_Final_v3.md
L3 — Tensor Volume	<u>Tensor- Volume- Layer-L3</u>	L3/paper/ + L3/L3_CLOSURE.md
L4 — Meta- Inference	<u>Meta- Inference- Layer-L4</u>	L4/paper/CAL_L4_Paper_v0.1.md

Collaboration: AMD-Instinct Labs — fa_dme on MI300X provides the empirical $O(n^2)$ baseline (measured: $n^{1.90}$, $R^2=0.996$) for the L4 Efficiency Hypothesis (§6.2).

v1.4 update note (June 2026). Since the v1.3 deposit, L3’s characterization phase closed. The headline result reorders the framework: **reconstruction fidelity $\neq$ causal conservation** — variance-optimal compression (Tucker) destroys the causal structure governance needs. CAL now commits to the validation order **Causality $>$ Topology $>$ Reconstruction**. A ground-truth-free causal metric (U) was validated, and causal conservation was shown to be **structural sparsity preservation** (an operator is causally conservative iff it preserves the observational invariants Ω_0 = reachability, coverage, consistency under compression). See §5.7 and the L3 closure document. This note records the evolution from the v1.3 open-problem framing; the body sections below are annotated where superseded rather than rewritten, preserving the citable v1.3 record.

Abstract

Thesis: scalable cognition requires hierarchical semantic compression. More precisely: inference scalability depends on *semantic state topology*, not raw context size. The quality of AI governance scales with the structure of the compressed state space — with whether the compression preserves the topology of governance-relevant states — not with the number of tokens processed. As AI systems generate outputs at scales that exceed human supervisory capacity, the fundamental challenge of AI governance shifts from generation to comprehension. We propose that this challenge cannot be solved by increasing human review bandwidth or improving artifact-level tooling, because both responses operate at the wrong level of abstraction. We introduce the **Cognitive Abstraction Layer (CAL) architecture**: a five-level hierarchy spanning raw token streams (L0) to autonomous meta-inference on compressed semantic volumes (L4). The core invariant of the architecture is *semantic conservation*: each layer must compress state while preserving sufficient causal and decisional structure for effective reasoning at the layer above. We formalize this requirement through the concept of **Semantic Information Density (SID)** — the quantity of decision-relevant cognitive structure preserved per unit of representation, as distinct from raw token count. The central claim is that the quality of AI governance scales with SID at each layer boundary, not with context window size — a separation of computational capacity from abstraction quality that current architectures conflate. L1 is instantiated by existing large language model inference, but is not formally structured as a governance layer; L2 is formalized and empirically validated by the companion paper TCO-L2 [Chancay 2026]; L3 and L4 represent open research directions with defined formal requirements and testable hypotheses.

Keywords: cognitive load, AI governance, semantic compression, tensor representation, hierarchical inference, human-AI collaboration, Semantic Information Density, Natural

1. Introduction

The dominant paradigm in AI-assisted software development positions the human as a validator of AI-generated artifacts. As agent pipelines mature, this paradigm is reaching a structural limit: the volume of output generated by concurrent AI agents systematically exceeds the human capacity for artifact-level review. Analysis of code repositories with high proportions of AI-generated content documents increasing defect rates despite larger output volumes [GitClear 2025]. The bottleneck is not generation — it is the human capacity to maintain systemic comprehension over the outputs of multiple concurrent agents operating at different stages of a shared pipeline.

Two common responses to this problem are: (1) reduce the human’s role further, replacing review with automated quality checks, and (2) improve tooling around the existing paradigm — better diffs, richer IDEs, faster search. We argue that both responses are incomplete, because they operate at the wrong abstraction level. The problem is not that humans review too little, nor that their tools are inadequate. The problem is that the information presented to human supervisors is at the wrong abstraction level for the cognitive operations they are being asked to perform.

This paper proposes a third response: a hierarchical architecture of semantic abstraction layers that progressively compresses AI-generated state into forms appropriate for increasingly autonomous inference. We call this the **Cognitive Abstraction Layer (CAL) architecture**. It has five levels, L0 through L4. The human supervisor appears at Level 2 — not as a reviewer of artifacts, but as an orchestrator of system states. Below Level 2, generation and local inference are automated. Above Level 2, governance inference is automated. At Level 2, the human operates at the **Natural Cognitive Frontier** [Chancay 2026]: the abstraction level where cognitive demand is calibrated to human capacity, decisions are expressible in natural language, and judgment produces policy rather than correction.

The key question the architecture poses is not “how much can humans supervise?” but “what is the minimum semantic representation that preserves the structure required for effective governance at each level?” This question is formalized through the concept of **Semantic Information Density (SID)**: the quantity of decision-relevant cognitive structure preserved per unit of representation. The thesis of the CAL architecture is that maximizing SID at each layer boundary — not expanding context windows — is the path to scalable AI governance.

The current paper is a research agenda. We do not report new experimental results for L3 and L4. We provide: (a) formal definitions of each layer and the properties required at each boundary, (b) a summary of the L2 empirical anchor (TCO-L2), (c) concrete open problems with testable hypotheses for L3 and L4, and (d) the SID metric as a unifying measurement framework. The goal is to establish the theoretical framing and priority of the full architecture before individual layer papers are submitted.

1.1 Scope and Non-Goals

CAL is a research architecture for AI pipeline governance in multi-agent software development contexts. It is not:

- **A theory of general cognition or AGI.** The word “cognitive” in CAL refers to the supervisory operations that agents and humans perform over pipeline state, not to cognition in the cognitive science or AGI sense. CAL makes no claims about how biological or artificial general intelligence operates.
- **A proposal to eliminate human governance.** L2 (TCO-L2) places the human orchestrator at the center of the architecture. L3 and L4 automate inference for decisions where human working memory is the bottleneck — they do not eliminate human accountability, which remains anchored at L2.
- **A general inference efficiency claim.** The L4 Efficiency Hypothesis applies

specifically to governance inference over semantically compressed pipeline state. It does not claim that hierarchical abstraction improves efficiency for arbitrary tasks — especially creative, open-ended, or low-structure domains where semantic compression may destroy rather than preserve relevant structure.

- **A complete, deployed system.** L3 and L4 have defined formal requirements and open problems, not implementations. Sections 5 and 6 describe what must be built and what must be proven, not what exists.

The scope is: multi-agent AI pipelines with measurable quality dimensions (as defined by ϕ in Section 4.1) and known pipeline structure. Extension to other domains requires domain-specific instantiation of ϕ and fresh validation of the L1→L2 interface in that domain.

2. The CAL Architecture

The CAL architecture is a five-level hierarchy. Each level receives compressed state from the level below, performs a defined transformation, and outputs a further-compressed representation to the level above. The architecture is bidirectional: state flows upward (L0 → L4), and policy flows downward (L4 → L0). At L2, the human orchestrator is the bidirectional interface: reading compressed governance signals upward from the tensor and injecting natural language policy downward into the pipeline.

```

L4 Meta-Inference Layer       $M(V) \rightarrow \{\text{decisions, predictions, adaptations}\}$ 
    ↑  $M: V \rightarrow \text{structured governance outputs}$ 
L3 Tensor Volume Layer       $V(T_1, T_2, \dots, T_n)$ 
    ↑  $C: \{T\} \rightarrow V$  [composition operator – open problem]
L2 Cognitive Tensor Layer     $T[d,i,j,k]$  ← TCO-L2
[Chancay 2026]
    ↑  $f \circ \phi: \{A\} \rightarrow T$     $I: T \rightarrow \{\Omega, \Delta, P, \Xi\}$ 
    ↓ Human orchestration at the NCF (natural language policy)
L1 Semantic Feature Layer    {embeddings, relations, quality signals}
    ↑ LLM inference: tokens → features [instantiated, not formalized as layer]
L0 Token / Artifact Space    raw AI-generated outputs
  
```

The core invariant at every layer boundary $L_k \rightarrow L_{k+1}$ is **semantic conservation**: the compressed representation at L_{k+1} must preserve sufficient causal and decisional structure for effective reasoning at L_{k+1} . A representation that fails this invariant destroys governance-relevant information and cannot be recovered at higher layers. Measuring the degree to which each boundary satisfies this invariant is the role of SID (Section 7).

3. L0–L1: Foundation Layers

3.1 L0: Token and Artifact Space

L0 is the space of raw AI-generated outputs: source code, configuration files, test results, documentation, deployment manifests, system logs. It is the output of AI generation agents (code agents, design agents, deploy agents) and the entry point to the abstraction hierarchy. L0 is not a contribution of the CAL architecture — it is the existing output of any AI-assisted development pipeline.

The key property of L0 is *volume asymmetry*: modern agentic pipelines generate artifacts at rates that exceed human review capacity. This is the fundamental motivation for the hierarchy. Without abstraction, L0 directly reaches the human supervisor, producing cognitive overload and the degraded judgment documented in [Sweller 1988, Parasuraman & Manzey 2010].

3.2 L1: Semantic Feature Layer

L1 is the space of semantic features extracted from L0 artifacts by inference models. This includes:

- **Embeddings:** dense vector representations of artifact semantics
- **Relational features:** dependency graphs, call graphs, architectural relationships
- **Quality signals:** static analysis outputs (cyclomatic complexity, security vulnerabilities, maintainability indices), test coverage, linting results

Critical clarification — L1 is instantiated, not formalized. Existing large language models generate L1-type features as intermediate representations in service of their primary generation and evaluation tasks. They do not expose a structured, persistent L1 state that formally serves as input to a governance layer. The CAL architecture does not propose building L1 from scratch — it formalizes the interface between L1 and L2, specifying what properties L1 outputs must have to support effective L2 governance.

Formally, L1 produces a set of per-artifact feature representations $F = \{f_1, f_2, \dots, f_n\}$, where each f_i is a local representation of a single artifact. L1 does not perform cross-artifact, cross-temporal, or cross-agent aggregation — these operations are structurally the responsibility of L2. The $L1 \rightarrow L2$ interface is therefore: a collection of per-artifact feature representations, sufficient in dimensionality and semantic content to support the ϕ vectorization function at L2.

4. L2: Cognitive Tensor Layer — TCO-L2

L2 is formalized and empirically validated by the companion paper TCO-L2 [Chancay 2026]. This section summarizes its contribution in the context of the CAL architecture. Full technical detail is in [Chancay 2026].

4.1 Vectorization: ϕ — L1 Features $\rightarrow$ Quality Vector

$$\phi: A \rightarrow V \in [0,1]^{11}$$

The vectorization function ϕ maps each artifact to an 11-dimensional quality vector $V = (v_1, \dots, v_{11})$. Dimensions span: functional correctness (v_1), architectural alignment (v_2), scalability projection (v_3), security risk (v_4 , inverted), observability coverage (v_5), testability (v_6), maintainability (v_7), technical debt (v_8 , inverted), performance (v_9), evaluator confidence (v_{10}), and anomaly score (v_{11} , inverted).

Epistemological note: v_1, v_2, v_3, v_9 are **supervisory estimators** — heuristic semantic signals generated by LLM assessment, without universal ground truth. v_4, v_6, v_7, v_8 are **verifiable ground truth** — outputs of deterministic static analysis (Bandit, Radon). This distinction is critical for the calibration of SID at the $L1 \rightarrow L2$ boundary.

4.2 Aggregation: f — Quality Vectors $\rightarrow$ Cognitive Tensor

$$f: \{V\} \rightarrow T[d,i,j,k] \in \mathbb{R}^{n \times |S| \times |A| \times |T_idx|}$$

The aggregation function f builds a four-dimensional tensor indexed by quality dimension (d), pipeline stage (i), agent (j), and time index (k). The tensor is the core data structure of L2: the shared index structure $[d,i,j,k]$ makes multidimensional supervisory operations first-class.

Two experimental scenarios demonstrate the operational necessity of this representation:

S3 — Gradual technical debt: per-cycle degradation in v_8 of -0.08 falls within individual evaluator noise ($\sigma=0.07$, SNR per cycle = 1.14). The 3-cycle cumulative delta (-0.24 , SNR = 3.43) reliably crosses the alert threshold. Monte Carlo simulation ($n=1000$) confirms: artifact-level detection rate 32.3% vs. tensor Δ detection rate 66.9% (+34.6 percentage point gap). The SQL equivalent requires a self-join reconstructing the coordinate system at the application layer. The tensor operation $T[v_8, :, :, \theta] - T[v_8, :, :, 3]$ is one line.

S5 — Inter-agent conflict: two agents produce opposing quality profiles ($v_4\Delta=0.65$, $v_6\Delta=0.50$). Per-artifact review requires the supervisor to hold both outputs in working memory simultaneously. The tensor P operation $|T[:, i, j_0, k] - T[:, i, j_1, k]|$ detects both conflicting dimensions automatically. SQL requires $N \times (N-1)/2$ pairwise joins for N

agents.

4.3 Inference: I — Tensor → Governance Outputs

$I: T \rightarrow \{\Omega, \Delta, P, \Xi\}$

Output	Description	Threshold
Ω	Global system state (stable / warning / critical)	$stable \geq 0.70$
Δ	Trend — early warning signal per dimension	$ slope > 0.05$
P	Systemic risk — inter-agent quality conflicts	$ diff > 0.30$
Ξ	Ranked recommendations by estimated $\partial\Omega/\partial action$	Top 3Δ + top $2P$

4.4 The Natural Cognitive Frontier (NCF)

The NCF is the abstraction level at which cognitive demand is calibrated to human capacity: decisions do not exceed working memory bounds ($\leq 4\text{--}5$ chunks, Cowan 2001), are expressible in natural language, and produce policy rather than artifact correction. At L2, the human orchestrator reads $\{\Omega, \Delta, P, \Xi\}$ and injects policy P_{new} in natural language.

NCF operationalization in TCO-L2 uses four observable proxies: NASA Raw-TLX subscales (working memory saturation), correction log consistency σ across fault categories, accuracy variance $\sigma(\text{accuracy})$ across tasks $T1\rightarrow T4$, and interaction timer IQR (attention fragmentation).

4.5 L2 as the Empirical Anchor for Semantic Conservation

The TCO-L2 experiment (between-subjects RCT, $n=40$ software engineers) tests whether participants can make more accurate governance decisions from T than from raw L0 artifacts. This is not only an HCI claim — it is the empirical validation that the $L0\rightarrow L2$ abstraction is semantically conservative.

The dual purpose of the experiment: (1) validate that the TCO-L2 dashboard reduces cognitive load and improves oversight quality (HCI contribution); (2) establish empirical evidence that tensor representation preserves sufficient decision-relevant structure for effective governance (CAL contribution). A positive result on H2 (Cohen’s $d > 0.50$ on detection accuracy) is the first measured data point for $SID(L0\rightarrow L2)$.

5. L3: Tensor Volume Layer

L3 is the first layer above human working memory in the CAL hierarchy. It operates on **tensor volumes** V — collections of L2 cognitive tensors aggregated across sessions, pipelines, or time windows — without requiring the human as an inference substrate.

5.1 Formal Definition

Let $T^{(s)}$ denote the cognitive tensor produced by session s . A tensor volume is:

$$V = C(\{T^{(s)}\}_{s=1}^{n})$$

where C is the **composition operator** — the central open problem of L3. C maps a collection of session tensors into a unified higher-order structure that preserves their causal and temporal relationships.

5.2 Requirements for the Composition Operator C

For V to be semantically conservative with respect to L4 inference, C must satisfy:

1. **Causal preservation:** causal relationships between quality dimensions recorded within any $T^{(s)}$ must be encodeable in V . Example: if a security hardening decision at $k=2$ caused a testability degradation at $k=3$ within session s , C must represent this dependency.
2. **Temporal coherence:** the time indices of constituent tensors must be composable into a global temporal ordering. Cross-session drift (e.g., progressive architectural degradation across 10 sprint cycles) must be representable in V .
3. **Dimensional stability:** the 11-dimensional quality vector structure must be preserved across the composition. L4 must be able to apply the same inference operations (Ω , Δ , P , Ξ) to slices of V that it applies to T .
4. **Tractability:** C must be computable for continuous pipelines. The size of V must not grow linearly with n for large n .

5.3 Tractability and Mitigation Strategies for Combinatorial Explosion

The composition operator C faces a combinatorial growth risk that must be treated as a primary design constraint, not a secondary concern. A naive tensor product $C = T^{(1)} \otimes T^{(2)} \otimes \dots \otimes T^{(n)}$ grows exponentially in dimensionality with n , directly violating the tractability requirement (Property 4 in Section 5.2) and negating the L4 Efficiency Hypothesis. Four mitigation strategies are candidates for the L3 research agenda:

1. **Sparse tensor representation.** Entries in V corresponding to agent-stage-dimension combinations that never co-occur in practice are structurally zero. Sparse formats (COO, CSR, block-sparse) reduce storage and computation to $O(k)$ in the number of nonzero entries k . If pipeline interactions are sparse — which is empirically likely given the bounded number of agents and quality dimensions — k grows sub-linearly with n .
2. **Low-rank approximation.** Tucker and CP decompositions [Kolda & Bader 2009] approximate high-dimensional tensors as products of smaller factor matrices. If the effective rank of V is bounded by the number of distinct governance-relevant pipeline patterns (rather than by n), low-rank V is tractable. This is plausible if pipelines exhibit a small set of recurring failure modes — a testable empirical hypothesis.
3. **Selective retention (dynamic forgetting).** Not all past $T^{(s)}$ contribute equally to current governance. C can implement a selective update rule — analogous to the gating mechanism in SSMS and LSTMs — that discards low-SID sessions and retains high-SID ones. This keeps the effective n bounded at any point in time.
4. **Manifold projection.** If the space of governance-relevant pipeline states lies on a low-dimensional manifold embedded in the full tensor space, V can be projected onto this manifold with bounded information loss. The **governance manifold hypothesis** — that the reachable space of pipeline quality states is low-dimensional — is the key assumption that would make manifold projection viable. Testing this hypothesis on the TCO-L2 corpus is a natural next step.

The combinatorial explosion risk does not invalidate L3; it defines the primary engineering constraint. Any candidate composition operator C must be evaluated against these four mitigation dimensions before its tractability can be claimed.

5.4 Open Research Questions at L3

- **Algebraic structure of C :** does C belong to a known algebraic family (tensor product, contraction, concatenation along a new axis, low-rank approximation)? What are the guarantees of each candidate?
- **Attractor states:** does V exhibit stable configurations that recur across sessions regardless of initial conditions? If so, these attractors may represent semantic invariants of the pipeline — structural patterns in agent behavior that persist across context changes.
- **Phase transitions:** does V exhibit discontinuous changes in global state (semantic

regime shifts) analogous to phase transitions in physical systems? Detecting the precursors of these transitions is a key L4 use case.

- **Scalability:** how does the effective rank of V grow with n ? Polynomial growth would suggest that $SID(L2 \rightarrow L3)$ decreases with pipeline scale — a critical failure mode to characterize.

5.5 Connections to Existing Research

State Space Models. SSMs such as Mamba [Gu & Dao 2023] demonstrate that temporal sequences can be compressed into fixed-size state representations with sub-quadratic complexity while preserving long-range dependencies via selective state spaces. The composition operator C in L3 is a semantic analog of the SSM state transition: it must compress temporal state while preserving structure. The key difference is that SSMs operate on raw token sequences; C operates on semantically structured quality tensors. Whether the selectivity mechanism of SSMs extends to the structured inputs of L3 is an open question.

Renormalization Group Theory. In statistical physics, renormalization reduces the degrees of freedom in a system while preserving macroscopic behavior [Wilson 1971]. L3 is a semantic coarse-graining: it reduces the dimensionality of the cognitive state space while preserving the macroscopic governance properties (trends, conflicts, anomalies) that matter for L4 inference. The renormalization group formalism may provide mathematical guarantees on what is preserved and what is lost under C , analogous to how fixed points of the RG flow characterize universality classes.

Tensor Decomposition. Tucker decomposition, CP decomposition, and tensor train/ring formats [Kolda & Bader 2009] offer tractable low-rank approximations of high-dimensional tensors. These may be candidates for C when exact composition is intractable.

Causal Representation Learning. The do-calculus requirement for C (Section 5.2, Property 1) connects directly to causal representation learning [Schölkopf et al. 2021]. Standard unsupervised representation learning preserves statistical correlations but may destroy causal identifiability — precisely the silent failure mode identified in Section 7.1, where acceptable contemporaneous SID coexists with destroyed temporal causality. Two methods from this field are directly applicable to L3: (a) **causal graph priors**, which encode known or hypothesized causal relationships between quality dimensions (e.g., security hardening $\rightarrow$ testability degradation) as structural constraints on C , preventing the composition from collapsing causal edges into spurious correlations; and (b) **causal regularization**, which adds a penalty term to the learning objective of C proportional to the degree to which interventional distributions $P(d_j | do(d_i))$ are not recoverable from V . The primary challenge is that causal discovery in compressed spaces is computationally expensive, especially when the true causal graph is unknown. Causal graph priors derived from domain knowledge about software pipeline dynamics are therefore the more tractable near-term approach.

5.6 The Governance Manifold Hypothesis

The four tractability mitigations in Section 5.3 are most effective when a single foundational assumption holds. We state this assumption explicitly as a testable hypothesis that may constitute the primary mathematical contribution of future L3 research:

Governance Manifold Hypothesis: the set of governance-relevant pipeline states is a low-dimensional manifold M_{gov} embedded in the full tensor space $\mathbb{R}^{m \times |S| \times |A| \times |T_{\text{idx}}|}$. The intrinsic dimensionality of M_{gov} is bounded by the number of distinct governance-relevant failure patterns the pipeline can produce — not by the full tensor dimensionality.

If true, the consequences propagate through the entire architecture:

- **Tractability:** V needs only to represent M_{gov} , not the ambient tensor space. The effective dimensionality of V is $\dim(M_{\text{gov}})$, which may be orders of magnitude smaller than the ambient dimension.
- **SID geometry:** $SID_D(Lk \rightarrow Lk+1) > \theta$ becomes a topological question — does C preserve the connectivity and local curvature of M_{gov} ? — rather than a high-

dimensional statistical estimation problem.

- **L4 Efficiency:** if $M(V)$ operates on M_{gov} , then $\text{Cost}(M(V)) = O(\dim(M_{\text{gov}}))$, decoupled from n . The L4 Efficiency Hypothesis is then a consequence of the manifold hypothesis, not an independent claim.
- **Semantic collapse framing:** collapse (Section 7.3) corresponds precisely to C projecting M_{gov} onto a lower-dimensional submanifold, destroying distinctions between failure modes that are topologically separated in M_{gov} .

Testability at L2. The hypothesis is testable on the TCO-L2 corpus without new data collection. If the quality vector trajectories across S1–S5 can be embedded in a low-dimensional space — using UMAP [McInnes et al. 2018] or Isomap — while preserving inter-scenario distances, this is empirical support for a low-dimensional governance manifold at L2. The dimensionality of this embedding is the first measurable proxy for $\dim(M_{\text{gov}})$, and provides the quantitative justification for whether low-rank and manifold projection strategies in Section 5.3 are warranted.

This positions topology — not compression ratio — as the central mathematical object of CAL. The right question is not “by how much did we compress?” but “did we preserve the topology of the governance manifold?”

[v1.4 refinement] L3’s results sharpen this further: the central object is **causal structure**, not topology *per se*. The governance manifold was confirmed low-dimensional ($\dim \approx 2\text{--}3$, trustworthiness ≥ 0.96) — but found to be *static* (reconstructible with the time axis averaged out) while causality is *temporal*. A purely topological criterion (preserve the manifold) can therefore be satisfied while causal structure is destroyed. The refined question is “did we preserve the **causal** structure?” — with topology subordinate to causality (see §5.7).

5.7 L3 Results — Causal Conservation under Compression (v1.4, June 2026)

This section summarizes the L3 characterization phase, which closed after the v1.3 deposit. It supersedes the “open problem” framing of §5.2–§5.4 for the specific question of *what causal preservation requires*; the formal requirements (Properties 1–4) stand, but their relative priority and the operative success criterion are now established empirically. Full record: L3/L3_CLOSURE.md.

The headline result — reconstruction $\neq$ causality. Low-rank Tucker decomposition (the §5.3.2 candidate) preserves 98% of variance and is tractable (Property 4 holds, $\kappa(V)$ sub-linear in n , $195.6\times$ compression) — yet it **destroys** the causal structure between quality dimensions. On a synthetic corpus with known causal ground truth, against a perfect raw control (causal F1 = 1.000), the Tucker reconstruction scored causal F1 = 0.135, fabricating spurious causal edges. Variance-optimal and causality-preserving compression are *different objectives*.

The reordering. CAL commits to a validation order: **Causality** > **Topology** >

Reconstruction. An operator must demonstrate causal conservation before geometric/topological fidelity, never the reverse. This promotes the §7.1 “silent failure mode” (acceptable contemporaneous SID with destroyed temporal causality) from a footnote to the framework’s primary constraint.

A ground-truth-free causal metric (U). Because no causal graph exists at deployment, causal conservation must be measurable without ground truth. We validated **U** — the correlation of off-diagonal PCMCi val-matrix flow — via a calibration test (it orders a family of compressions by their known causal fidelity, and collapses when causality is destroyed at fixed marginals). **U** is both a validated *instrument* and a usable optimization *objective*. A key negative result: causal fidelity is carried by **discrete structure**, not continuous magnitudes — differentiable magnitude proxies fail to reproduce **U**’s ordering.

The operator and the operative property. C is built as $C = C_{\text{causal}} \circ C_{\text{compress}}$ (Tucker compresses; a causal step preserves structure). The success criterion that closed L3:

An operator is *causally conservative* iff it preserves the observational invariants $\Omega = (R, C, S)$ — reachability, coverage, consistency — under compression, not reconstruction fidelity.

Tucker’s failure mode was diagnosed precisely: it preserves coverage (recovers true edges) and consistency (no sign errors) but **fabricates spurious edges** (reachability and edge count explode). A structural prune to the causal support recovers **75% of the causal-conservation gap with no ground truth** (U: 0.441 $\rightarrow$ 0.862; the deployment prune equals the ground-truth-oracle prune).

Consequence for the Governance Manifold Hypothesis (§5.6). $\dim(M_{\text{gov}}) \approx 2-3$ is confirmed, but M_{gov} is *static*; whether a low-dimensional **causal** manifold exists is reframed as an open L4 question. The manifold is retained as a descriptor, not as the projection that defines C.

Status. L3 characterization is $\sim 95\%$ closed; the residual (the gap from U=0.862 to 1.0) is a declared frontier (flow magnitude / higher-order structure), not a defect. The composition operator delivered to L4 is the dual representation $V = (V_{\text{Tucker}}, G_{\text{pruned}})$ with $\kappa(V)=1296$ — closing condition (a) of the L4 Efficiency Hypothesis (§6.2).

6. L4: Meta-Inference Layer

L4 performs inference on tensor volumes V rather than on individual tensors T or artifacts A . It is the fully automated inference layer of the CAL architecture: no human working memory is required as the inference substrate.

6.1 Formal Definition

$M: V \rightarrow \{\text{decisions, predictions, adaptations}\}$

The meta-inference function M operates on compressed semantic volumes. Because V (by the properties required of C) encodes causal relationships, temporal coherence, and dimensional stability, M can perform reasoning that is not possible at lower layers:

- **Causal discovery:** identifying which quality dimensions causally influence others, across sessions. Example: establishing that security hardening interventions systematically reduce testability with a lag of 1–2 cycles.
- **Drift prediction:** predicting future quality trajectories from historical volume dynamics, before the signal crosses the threshold detectable at L2.
- **Cross-pipeline conflict detection:** identifying conflicts between architectural strategies that span multiple sessions or pipelines.
- **Policy generation:** producing governance policies from historical patterns, which can be passed downward to L2 as default policy suggestions for human review or — in high-confidence, low-stakes cases — injected directly.

6.2 The L4 Efficiency Hypothesis

The fundamental motivation for the CAL hierarchy is that semantic compression reduces inference cost. This is formalizable as:

L4 Efficiency Hypothesis: there exists an inference architecture such that the cost of $M(V)$ scales with $\kappa(V)$ — the structural complexity of V (e.g., its effective rank, the entropy of its attractor distribution, the size of its causal graph) — where $\kappa(V)$ grows significantly more slowly than $O(n^2)$ in n (the raw artifact count of the underlying L0 space).

This is not a claim that meta-inference is exponentially cheaper than token-level attention in all cases. It is a claim that if semantic abstraction is semantically conservative, inference cost can be decoupled from the volume of the original artifact stream and coupled instead to the structural complexity of the compressed governance state. This hypothesis connects to the observation that current transformer architectures re-evaluate redundant relationships at every forward pass [Tay et al. 2022]; a hierarchy of semantic abstractions would

eliminate this redundancy by compressing it away before inference.

Empirical testing of this hypothesis requires: (a) a concrete definition of C (the composition operator) and $\kappa(V)$ (the structural complexity of V), (b) a comparison of $M(V)$ cost vs. equivalent flat-context inference on raw artifacts, and (c) measurement of governance accuracy under both approaches. TCO-L2 provides the L2 baseline for (c).

[v1.4 status] Conditions (a) and (b) are now **met** (June 2026). (a): L3 closed with $C = C_{\text{causal}} \circ C_{\text{compress}}$ delivering $\kappa(V)=1296$ (effective rank of the Tucker core, 195.6 $\times$ compression) — the first of the three $\kappa(V)$ candidates above, made concrete (§5.7). (b): AMD-Instinct measured the flat-context $O(n^2)$ curve on MI300X at $n^{1.90}$ ($R^2=0.996$, confirmed quadratic). The κ vs n^2 cost contrast is therefore runnable on hardware; condition (c) — governance accuracy of $M(V)$ vs flat-context — remains the open gate (RCT-adjacent). The hypothesis is **not yet proven**: (a)+(b) make it *testable*, not *confirmed*.

6.3 Open Research Questions at L4

- What is the minimal V that supports M for a given class of governance decisions?
- Can M be trained end-to-end from labeled governance outcomes, or does it require explicit causal structure encoded in V as supervision?
- What are the failure modes of M when C fails to preserve causal structure? How does governance accuracy degrade?
- At what scale (number of agents, sessions, quality dimensions) does $M(V)$ outperform equivalent flat-context LLM inference on governance tasks?

6.4 The Representational Convergence Conjecture

The L4 Efficiency Hypothesis (Section 6.2) treats the compressed semantic substrate as something CAL constructs *post hoc*, by reading the outputs of an unmodified generator. A stronger — and, if true, more consequential — possibility is that the compression need not be *post hoc* at all. We state it as an explicit, falsifiable conjecture, separate from the architecture’s near-term claims:

Representational Convergence Conjecture (RCC): the representation that is *governance-optimal* (minimal sufficient for human-supervisable causal observability, i.e. CAL’s V at high SID) and the representation that is *compute-optimal* (minimal sufficient internal state for the generator to continue producing semantically coherent output without re-deriving the full context at each step) converge to the same object — or to two objects related by a tractable, possibly learned, translation.

Motivation. Current transformer inference pays a cost that scales with the *raw volume* of context ($O(n^2)$ attention), not with its *structural complexity*: a 100k-token context describing a structurally simple state costs the same as one describing a complex one. Section 6.2 already observes that this re-evaluates redundant relationships at every forward pass. The RCC pushes the observation one step further: if a semantically conservative compressed state V exists and persists, a generator coupled to it would not need to regenerate semantic density token-by-token — it would only need to emit *deltas* against the persistent state and let an update operator fold them in. The generator’s compute migrates from “regenerate everything” to “update the compressed state.” Under the RCC, the substrate that does this folding is *the same substrate CAL maintains for human governance*.

Why this is distinct from L2. L2 controls the *transformation of the pipeline* — what the agentic system produces and how — via natural-language re-orchestration. It does **not** govern the generator’s compute. The RCC concerns the *transformability of the computation itself*: whether the generator can be reorganized to exploit the same compressed state. These are the same word-root operating in two planes; L2’s plane is established, the compute plane is the conjecture. Conflating them would overstate L2’s claims, so the architecture keeps them separate: **L2 is governance over pipeline transformation; the RCC is a hypothesis about compute-side transformability at L3–L4.**

The hard requirement. The conjecture’s difficulty is concentrated in one place: CAL’s ϕ is today a *non-differentiable, human-legible* compressor (static analysis + discrete LLM-

judge). For the RCC to hold, there must exist a ϕ' that is *also* couplable to the generator's gradient or latent dynamics — a compressor that is simultaneously legible to a human supervisor and usable as (or translatable to) the generator's own latent state. Whether such a ϕ' exists, and whether the human-optimal and compute-optimal compressions actually coincide rather than merely correlate, is open. The negation is entirely plausible: the two pressures (legibility vs. predictive sufficiency) may pull toward different manifolds.

Falsifiability. The RCC predicts that, for a generator coupled to a persistent semantic state, (a) inference cost decouples from raw context length and couples to $\kappa(V)$ (the L4 Efficiency Hypothesis as a special case), and (b) a single learned translation suffices to map between the governance state V and the generator's update state, with bounded reconstruction loss on both governance accuracy *and* generation quality. A failure of (b) — where minimizing governance-SID and minimizing generation loss require provably distinct representations — refutes the strong form and leaves only the post-hoc reading of Section 6.2.

Relation to existing lines. The RCC sits at the intersection CAL already gestures at: State Space Models (Mamba, S4) make the “persistent compressed recurrent state” computationally concrete but do not require that state to be human-legible; the Governance Manifold Hypothesis (Section 5.6) makes the governance state low-dimensional but does not require it to be the generator's state. The RCC is the conjecture that these two low-dimensional objects are the same — that *the cognitively-optimal representation for a human supervisor and the computationally-optimal representation for the generator are one representation*. If it holds, a single state buys both cheap human oversight and cheap inference; the result would not be “a new kind of transformer” in the sense of a modified attention mechanism, but a transformer coupled to an external, persistent, legible semantic substrate — closer to a state-space architecture with a governance-readable state.

This conjecture is the *north star* that justifies building L2 well: it is the reason a governance-grade compressed representation may turn out to matter far beyond governance. It is explicitly **not** a claim of the CHI 2027 / Paper 1 submission, which is anchored at L2.

7. Semantic Information Density: A Unifying Metric

The CAL architecture requires a metric that can be applied at every layer boundary to assess whether the abstraction is semantically conservative. Without such a metric, the claim that each layer “preserves decision-relevant structure” is unfalsifiable. We propose **Semantic Information Density (SID)** for this purpose.

7.1 Formal Definition (Preliminary)

Let R_k denote the representation at layer L_k , and let D be the set of governance decisions relevant to the pipeline (e.g., deploy / halt / escalate / re-orchestrate). Define the **decision-relevant information content** of R_k as the mutual information between R_k and D :

$$I_{\text{decision}}(R_k) = H(D) - H(D \mid R_k) = I(D ; R_k)$$

i.e., the reduction in uncertainty about governance decisions D given the representation R_k . SID at a layer boundary is then:

$$\text{SID}(L_k \rightarrow L_{k+1}) = I_{\text{decision}}(R_{k+1}) / I_{\text{decision}}(R_k) \in [0, 1]$$

A SID of 1.0 means R_{k+1} is as informative as R_k for governance decisions. A SID below threshold θ means that compression has destroyed governance-relevant information that cannot be recovered at higher layers.

SID differs from Shannon entropy in that it is conditioned on the governance decision set D , not on information content in the abstract. A highly entropic representation can have low SID if its entropy is dominated by noise irrelevant to D .

Causal structure. A critical dimension of SID not captured by mutual information alone is causal structure preservation. We say a representation R_k preserves the causal structure of its source if, for any pair of quality dimensions (d_i, d_j) where d_i causally produces d_j in

the underlying pipeline, this relationship is recoverable from R_k . Using Pearl's do-calculus notation, the conditional interventional distribution $P(d_j | \text{do}(d_i))$ must remain estimable from R_k . A boundary where this fails may still have acceptable contemporaneous SID while producing systematic errors in drift prediction and temporal governance — a silent failure mode.

Mathematical guarantee: $SID_D \in [0,1]$. By the data processing inequality [Cover & Thomas 2006], for any Markov chain $D \rightarrow R_k \rightarrow R_{k+1}$ — where R_{k+1} is produced by applying C to R_k — the mutual information satisfies $I(D; R_{k+1}) \leq I(D; R_k)$. This guarantees $SID_D \leq 1$ for any valid compression, without additional assumptions. $SID_D = 1$ (lossless) and $SID_D = 0$ (complete governance information loss) are the formal bounds. A direct consequence is that information destroyed at a layer boundary cannot be recovered at higher layers — making SID_D a strict, irreversible measure of information loss. Every compression either preserves governance information or loses it permanently.

SID is contextual, not absolute. Because $I(D; R_k)$ depends on the governance decision set D , SID is not a single scalar property of a layer boundary — it is a family of measurements indexed by governance class. We denote this explicitly as SID_D :

$$SID_D(L_k \rightarrow L_{k+1}) = I(D; R_{k+1}) / I(D; R_k) \quad \text{for governance class } D$$

Different governance classes yield different SID values for the same compression: a C that preserves deploy/halt decisions well may destroy re-orchestration signal entirely. Comparability of SID across domains (software pipelines, scientific research, financial modeling) requires either a shared $D_{\text{universal}}$ — likely intractable — or domain-specific declarations of D with explicit scope. All SID claims in this paper are relative to the software development governance class defined by the TCO-L2 experimental scenarios (S1–S5). Extension to other domains requires fresh operationalization of D .

The central open problem. The fundamental difficulty in operationalizing SID is measuring I_{decision} . The governance decision set D is not fixed: it depends on the class of pipeline tasks, the organizational context, and the severity calibration used. At L2, the TCO-L2 experiment provides an empirical proxy via governance accuracy α_2 on a finite set of labeled scenarios. At L3 and L4, there is currently no established methodology for measuring I_{decision} without a human reference signal. Defining I_{decision} rigorously — in a way that is independent of the specific evaluator and generalizes across pipeline types — is the primary mathematical open problem of the SID framework.

7.2 Operationalization at L2: The TCO-L2 Experiment

The TCO-L2 experiment provides the first operational measurement of SID. Let α_2 denote the governance accuracy of the experimental group (operating at L2) and α_0 denote the governance accuracy of the control group (operating at L0). A bounded estimate:

$$SID_{\text{empirical}}(L0 \rightarrow L2) \approx \alpha_2 / \max_possible_a$$

where $\max_possible_a$ is the accuracy achievable with perfect information. The experiment design anchors this with ground truth labels for all five experimental scenarios (S1–S5), making α_2 and α_0 directly measurable. H2 (Cohen's $d > 0.50$) is equivalent to asserting that $SID(L0 \rightarrow L2)$ is substantially above the baseline.

7.3 The Open Problem: SID Without a Human Reference

At L2, the human provides the calibration signal: human governance accuracy is the ground truth for SID measurement. At L3 and L4, this signal is unavailable — by definition, these layers operate without human working memory as the reference. The central methodological challenge of L3–L4 research is therefore:

How do we measure $SID(L2 \rightarrow L3)$ and $SID(L3 \rightarrow L4)$ without a human reference for governance accuracy?

Three candidate approaches:

1. **Downstream outcome tracking:** measure whether governance decisions produced by $M(V)$ lead to measurable pipeline improvements (defect reduction, deployment

stability) over time. This requires longitudinal deployment.

2. **Synthetic benchmarks with known causal structure:** generate pipelines with known ground truth causal graphs and test whether $M(V)$ recovers the correct causal relationships. This provides a controlled measurement of $C(V)$ and SID.
3. **Comparison against L2 human decisions on held-out cases:** for a set of novel governance scenarios not seen during training, compare $M(V)$ decisions against expert human decisions at L2. This uses the L2 human as a delayed reference signal.

Semantic collapse as a named failure mode. Beyond the measurement problem, a structural risk deserves explicit naming: **semantic collapse** occurs when a composition operator C preserves acceptable SID_D for each individual governance class D while simultaneously destroying the diversity of the governance state space — collapsing distinctions between states that are relevant to different decision classes. This is the governance analog of mode collapse in generative models: local fidelity is maintained, global coverage is lost. Semantic collapse is particularly dangerous because it passes contemporaneous SID checks while producing systematic failures on novel or rare governance scenarios.

Three mitigations are candidates for L3 research: (a) **entropy preservation constraints**, requiring that the marginal entropy $H(V)$ remains within a specified band of $H(T)$ so that V cannot become arbitrarily concentrated; (b) **attractor separation metrics**, requiring that distinct attractor states in T — corresponding to distinct pipeline failure modes — remain metrically separated in V above a minimum distance threshold; and (c) **governance diversity benchmarks**, test suites that deliberately include rare and distributional-tail failure modes, designed to detect representations that collapse rare-scenario coverage.

The governance manifold hypothesis (Section 5.6) gives semantic collapse a geometric interpretation: collapse is equivalent to C reducing the intrinsic dimensionality of M_{gov} below the number of topologically distinct failure modes, destroying the separability that makes governance decisions possible.

8. Research Agenda

Layer	Repo	Status	Key open problem	Testable hypothesis	Target venue
L0	—	Production	None (baseline)	—	—
L1	—	Instantiated by LLMs	Formal L1→L2 interface specification	LLM quality signals have $SID > 0$ for L2 ϕ	—
L2	TCO	Working paper v3.0 · platform deployed	NCF operationalization, ϕ calibration	H2: Cohen's $d > 0.50$; $SID(L0 \rightarrow L2) > 0.70$ Tucker preserves causal graph	CHI 2027
L3	Tensor-Volume-L3	Characterization closed (~95%) · see §5.7, L3_CLOSURE.md	Causal conservation = structural sparsity preservation (Ω); residual 25% = frontier	$\rightarrow$ refuted; <i>structural prune recovers 75% of causal headroom, no ground truth</i>	NeurIPS / ICML
L4	Meta-Inference-L4	gate-C closed · L4-A validated · κ vs n^2 contrast active	Production M(V); single-V operator (L4-B)	Cost(M(V)) = $O(\kappa(V))$ where $\kappa(V) \ll O(n^2)$; cond. (a)+(b) met, (c) open	NeurIPS / ICML
SID	Cross-layer	Partially defined (L2 only)	Measurement without human reference for L3–L4	SID degrades gracefully with compression rate	Cross-layer

Collaboration — AMD-Instinct Labs (L4):

- **Rol 1 (active):** `fa_dme seqLen` sweep on MI300X — empirical $O(n^2)$ baseline for condition (b) of the L4 Efficiency Hypothesis
- **Rol 2 (post gate-C):** `fa_dme` as kernel proxy for M(V); `probe_mfma_mapping.hip` for RCC empirical grounding
- Gate: L3 composition operator C must be validated before Rol 2 begins

Dependency chain: L3 gate-C $\rightarrow$ L4 Rol 2 $\rightarrow$ L4 Efficiency Hypothesis test (synthetic) $\rightarrow$ RCC empirical probe

9. Positioning Within Existing Research

The CAL architecture connects to several active research areas that it does not replicate but draws from:

AI Observability / MLOps: Datadog, New Relic, OpenTelemetry monitor operational health (latency, errors, resource utilization). TCO-L2 monitors semantic health (quality, debt, architectural alignment, inter-agent conflicts). CAL extends this to hierarchical governance.

Hierarchical Reinforcement Learning: HRL decomposes long-horizon tasks across temporal abstractions (options framework). CAL applies analogous hierarchical abstraction to governance, not to task execution.

State Space Models: SSMS [Gu & Dao 2023] compress temporal sequences with sub-quadratic complexity. L3 requires a semantic analog of the SSM state transition for

structured quality tensors.

Cognitive Systems / Global Workspace Theory: GWT [Baars 1988] proposes a shared workspace for integrating specialized cognitive processes. The L2 tensor is a computational analog of the global workspace: a shared index structure that integrates quality signals from multiple specialized evaluation agents.

Situation Awareness: Endsley's SA model (perception → comprehension → projection) maps to L0 → L1 → L2 in the CAL hierarchy. TCO-L2 is an operationalization of SA Level 2–3 for AI pipeline supervision.

10. Conclusion

The CAL architecture proposes a reframing of the AI governance problem: not as a question of how much humans can supervise, but as a question of what abstraction level at each layer boundary maximizes Semantic Information Density for governance decisions. TCO-L2 is the empirical anchor of this framework — it validates, via controlled experiment, that the L0→L2 abstraction is semantically conservative enough for effective human orchestration at the Natural Cognitive Frontier. That validation is the theoretical foundation for L3 and L4: if a human can govern effectively from a tensor representation, an automated inference layer operating on tensor volumes can do so without the human working memory constraint.

The central open problems — the composition operator C for L3, the measurement of SID without human reference, the L4 Efficiency Hypothesis — define a coherent research agenda with concrete testable hypotheses at each step. The claim is not that these problems are solved, but that they are the right problems to solve: progress on each one directly advances scalable AI governance.

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This paper establishes the theoretical framing and research agenda for the CAL architecture. Experimental results for L2 are reported in the companion paper TCO-L2 [1]. L3 and L4 are open research directions with defined formal requirements and testable hypotheses.